AARADHYA DEV TAMRAKAR

Kathmandu, Nepal • aaradhyadevtmr@gmail.com • +977 9844602050
LinkedIn • GitHub • Portfolio

PROFILE

Electronics, Communication and Information Engineering student at IOE, Kathmandu Engineering College (Year IV/I). Focused on Artificial Intelligence, robotics, and embedded systems. Experienced building ML pipelines (classification, regression, tree-based ensembles, agentic Text-to-SQL systems), deploying models via FastAPI and Docker, and developing Android applications with Kotlin/Jetpack Compose. Currently a Fellow at the Fusemachines AI Fellowship (2026, Wk 10/24 ongoing) and NSSR DataCamp Fellow (Cohort 2, 2026), while leading the ESP32-S3 edge-AI wearable fall-detection system SPARK as major project. Interested in applying AI to real-world sensor systems, robotics control, and edge intelligence.

EDUCATION

Bachelor of Engineering (B.E.) *Expected: January 2027*

Electronics, Communication and Information Engineering

Institute of Engineering (IOE), Kathmandu Engineering College | Year IV / Part I

Coursework: Machine Learning · Signals and Systems · Digital Electronics · Microprocessors · Control Systems · Communication Systems · DBMS · Project Management · Propagation & Antenna

EXPERIENCE

Vice Chair *2026 – Present*

IEEE KEC KTM Student Branch

- **Strategic planning:** Spearheading high-level decision-making to scale technical initiatives and member engagement.
- **Institutional relations:** Directing cross-functional executive committees and managing the broader engineering community ecosystem.

Event Manager *2026 – Present*

Electronics Project Club (EPC), Kathmandu Engineering College

- Managing logistics and coordination for engineering events and technical workshops.

Makerspace Ambassador *June 2026 – Present*

KEC Maker's Space, Kathmandu Engineering College

- Web development: Shipping site-wide fixes to the KEC Maker's Space website — image tag repair, content consistency, and UI improvements. Grants: free 3D printing, laser cutting, materials, and 1 Gbps WiFi access.

Vice Secretary *2025 – 2026*

IEEE KEC KTM Student Branch, Kathmandu Engineering College

- **Operations:** Managed official documentation, meeting minutes, and internal communication frameworks, streamlining branch operations.
- **Event delivery:** Orchestrated logistics for technical workshops and IEEE programs, significantly increasing student participation.
- **Volunteer management:** Supported the executive committee in coordinating volunteers for seamless event execution.

Resource Manager *2024 – 2026*

Electronics Project Club (EPC), Kathmandu Engineering College

- **Logistics:** Managed scheduling and resource allocation for engineering workshops and technical events across multiple departments.

FELLOWSHIPS & RECOGNITIONS

Fuse AI Fellow — Fusemachines AI Fellowship 2026 *Active*

Fusemachines

- **Admission:** Passed entrance examination (March 2026) covering linear algebra, calculus, probability, Python, and ML.

- **REST API:** Built a containerized customer REST API: FastAPI + PostgreSQL + Docker (4-layer architecture, asyncio concurrency).
- **Text-to-SQL agent:** Developed a five-stage agentic pipeline (Planner → Generator → Validator → Executor → Summarizer) achieving 100% execution success and 100% accuracy on a 50-question benchmark.
- **Churn & CLV modeling:** End-to-end pipeline with Logistic Regression / Ridge / SGD classifiers; Stratified 5-fold CV ROC-AUC 0.841 ± 0.005; threshold-tuned for top-200 high-risk segment; Papermill automation.
- **Tree-based ensemble:** Random Forest + XGBoost with ImbPipeline/SMOTE (zero leakage), SHAP explainability, GridSearch tuning, and Joblib serialization.
- **Probabilistic models:** Bayesian estimation (MLE/MAP/full Bayes), Dirichlet-multinomial inference, Gaussian process regression, probabilistic graphical models, and Bayesian logistic regression via PyMC/ArviZ/pgmpy.
- **Customer segmentation:** K-Means, Hierarchical (Ward/Complete/Average/Single dendrograms), and DBSCAN on UCI Online Retail II (~500K transactions); RFM + category-ratio feature engineering; Silhouette / Davies-Bouldin / Calinski-Harabasz validation; business narrative with executive summary.
- **Forecasting:** Benchmarked 9 classical-to-modern forecasters (SARIMA, Holt-Winters, Prophet, LightGBM, LSTM, XGBoost) on monthly S&P 500 data; built a 4-model ensemble (Holt-Winters + SARIMA + LSTM + XGBoost) outperforming all single models (MASE 2.44); confirmed via Diebold-Mariano significance test (p = 0.0092).

NSSR DataCamp Fellow — Cohort 2 *2026 – Present*

Nepalese Society of Student Researchers

- **Selection:** Competitively selected (April 2026) for a sponsored DataCamp Premium license; one of a limited cohort of Nepalese student researchers.
- **GenAI applications:** Engineered 6 applied GenAI solutions across product strategy and data workflows — deploying advanced prompt engineering and LLM constraint handling.
- **Data automation:** Reduced manual data preparation overhead by applying generative AI for automated null filtering, deduplication, and schema normalization.
- **Certifications:** SQL Associate + Python Data Associate (in progress); covering Data Manipulation in PostgreSQL, Pandas, Statistical Thinking, and ML with scikit-learn.

PROJECTS

SPARK — Wearable Fall Detection System (Major Project)

ESP32-S3 · TensorFlow Lite Micro · SHAP · MQTT · FastAPI · PostgreSQL

- **Architecture:** Two-layer edge AI on ESP32-S3 — fast kinematic threshold gate (<5ms), then an INT8 TFLite Micro 1D CNN confirmation layer (<100ms) — with per-event SHAP explainability computed at a local, non-cloud gateway; a preliminary literature pass found no 2024–2025 fall-detection work combining true MCU-class hardware with local-gateway SHAP specifically, though a formal database check is still pending.
- **Status:** Proposal submitted (signed hardcopy, July 2026) and successfully defended — panel optimistic about the project. Demo target: March 2027 (8th-semester boards).
- **Role:** AI & Backend work package (WP 1.0) — CNN training pipeline, FastAPI/PostgreSQL gateway; 4-person team, supervised by Er. Dipen Manandhar.

Text-to-SQL Agentic Pipeline

Python · FastAPI · Streamlit · GPT-4o-mini · PostgreSQL

- **System design:** Production-grade automated Text-to-SQL over classicmodels PostgreSQL for a 50-question benchmark.
- **Task 3:** Streamlit chat UI + vanilla Python prompt chaining with SQL safety validation (blocks DML/DDL) and query execution logging to JSON.
- **Task 4:** FastAPI modular planner-generator-validator-executor-summarizer state machine, up to 3 retries. 100.0% execution success and 100.0% accuracy across all 50 questions with zero retries.

Telco Churn Tree-Based Ensemble Pipeline

Python · XGBoost · Random Forest · SHAP · ImbPipeline · Joblib

- **Pipeline:** End-to-end classification and regression on Telco Customer Churn (7,043 rows, ~27% positive rate); exposed Accuracy Trap with naïve baseline.
- **Leakage prevention:** ImbPipeline restricts SMOTE to training folds only; ColumnTransformer for mixed dtypes.
- **Explainability:** SHAP global summary + local waterfall/force plots; retention recommendation for specific at-risk customer.

Gesture-Controlled Self-Balancing Robot (GCSBR) — Minor Project

Computer Vision · Arduino · MPU6050 · Stepper Motors · Android · MATLAB

- **Evaluation:** Described as 'major project level' by examiner.

- **Firmware V6.9.1:** Watchdog timer, atomic writes, differential ramp, integrator bleed, PID stabilization. MATLAB simulation → hardware deployment pipeline.
- **Interface:** Android gesture-control app (MediaPipe, CameraX, HC-05 BT); dual-hand gesture control; DRV8825 drivers on CNC Shield for real-time actuation.

Edge AI Stability Detection System

Python · Scikit-learn · FastAPI · Joblib

- ML system predicting platform stability from simulated IMU sensor data (tilt_x, tilt_y, angular velocity). 10,000-sample synthetic dataset; Random Forest — 99.8% test accuracy. REST API via FastAPI for real-time predictions.

Alpha Android Super-App

Kotlin · Jetpack Compose · Material3 · DataStore · Apache POI

- Modular Android super-app (SDK 36): SBR gesture control, multi-mode calculator (STD/PROG/LOGIC), budget tracker with eSewa Gmail parser, custom Astronomus font, full Material3 design system.

SysOptimizer — Windows Optimization Tool

Python · CustomTkinter · PyInstaller · WMI · PowerShell

- Standalone .exe with animated ring gauges, sparklines, process table. v4/v5: Power Plan switcher, RAM flush, Background Bloat panel (14 services/tasks), Startup Scanner.

Custom Processor FSM Design

VHDL · Vivado 2023.2 · FSM · Datapath · FPGA

- VHDL implementation of custom processor datapath and FSM supporting GCD and exponentiation operations. Simulated and verified in Vivado 2023.2.

Probabilistic Modeling Suite (Fusemachines Week 6)

Python · PyMC · ArviZ · pgmpy · scikit-learn · Pandas

- End-to-end Bayesian analysis suite: MLE/MAP/full Bayes estimation, sequential belief updating, Dirichlet-multinomial inference, multivariate Gaussian conditioning, probabilistic graphical models, Gaussian process regression, and PyMC Bayesian logistic regression with ArviZ posterior diagnostics. Artifact: telco_bayes_lr_v1.pkl.

Customer Segmentation Pipeline (Fusemachines Week 7)

Python · scikit-learn · scipy · Pandas · NumPy · Matplotlib · Seaborn

- Market segmentation on UCI Online Retail II (~500K transactions): RFM + extended features (AvgBasketSize, UniqueProducts, ReturnRate) + category spend ratios (keyword-bucketing). K-Means (Elbow + Silhouette), Hierarchical (Ward/Complete/Average/Single dendrograms), DBSCAN (k-distance ε tuning, ≥3 param combos). Validation: Silhouette / Davies-Bouldin / Calinski-Harabasz. Business narrative with cluster profiles and executive summary.

S&P 500 Forecasting Pipeline (Fusemachines Week 8)

Python · statsmodels · Prophet · LightGBM · TensorFlow/Keras · XGBoost · scikit-learn

- Benchmarked 9 forecasters (SARIMA, Holt-Winters, Prophet, LightGBM, LSTM, XGBoost, naive baselines) on monthly S&P 500 data (1990–2024). ACF/PACF diagnostics, residual analysis, error-by-horizon breakdowns. Built a 4-model ensemble (Holt-Winters + SARIMA + LSTM + XGBoost) achieving best MASE (2.44) and RMSE (96.7), beaten model confirmed significant via Diebold-Mariano test (p = 0.0092). Investment-memo-style recommendation: deploy ensemble, retain Holt-Winters as interpretable regulatory fallback.

Steel Surface Defect Classifier (Fusemachines Week 9)

Python · PyTorch · torchvision · Optuna · scikit-learn

- CNN classifier for the NEU surface-defect dataset (6 classes, 1,800 images) covering manufacturing quality-control scenarios; progressed through baseline network, convolutional model, data-augmentation hardening, and grid-search/Optuna hyperparameter tuning stages. Submitted July 2026.

Nexus — Personal AI Operating System

React (Vite) · FastAPI · SQLite + FTS5 · httpx · asyncio

- Project-centric AI operating system unifying Groq (Llama 3.3 70B) and Gemini into a single workspace via parallel fan-out routing (asyncio.gather). FastAPI backend + React frontend + SQLite + FTS5 context injection; multi-key rotation; project workspaces with persistent history. v2 complete: dark editorial UI, Google account routing sidebar.

SKILLS

Programming: Python, C, C++, Kotlin, SQL (PostgreSQL / SQLite), VHDL

Machine Learning: scikit-learn, NumPy, Pandas, XGBoost, Random Forest, Logistic/Ridge/SGD Classifiers, Lasso, ElasticNet, SMOTE (ImbPipeline), SHAP, GridSearchCV, Stratified K-Fold CV, Joblib, PyMC, ArviZ, pgmpy, Bayesian Inference (MLE/MAP/full Bayes), Gaussian Process Regression, K-Means, Hierarchical Clustering, DBSCAN, Silhouette/Davies-Bouldin/Calinski-Harabasz Validation, Time Series Forecasting (SARIMA, Holt-Winters, Prophet, LightGBM, LSTM)

Android Development: Kotlin, Jetpack Compose, Material3, DataStore, CameraX, MediaPipe, Apache POI

Deployment: FastAPI, REST APIs, Docker, Docker Compose, PostgreSQL (containerized), Streamlit, asyncio, Papermill, Joblib

Linux/Ubuntu: Ubuntu 26.04 LTS, dual-boot configuration, GRUB troubleshooting, apt package management, system diagnostics

Embedded Systems: Arduino, ESP32-S3, MPU6050, CNC Shield, DRV8825, Stepper Motors, HC-05 BT, UART, MQTT, FPGA, Vivado 2023.2

AI / Local LLM: Ollama, AnythingLLM, ipex-llm, DeepSeek R1 7B (Intel Arc iGPU), Prompt Chaining, Agentic Query Systems

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, MATLAB, LaTeX / Overleaf, SSMS

Other: Event Coordination, Project Management, Public Relations, Microsoft Office, Canva, Blender (Beginner)

TECHNICAL ACTIVITIES & CERTIFICATIONS

- IEEE WIE Nepal Section LaTeX Workshop 2026 — KEC Makerspace (May 5–6, 2026). Certificate of Participation. Skills: LaTeX, Technical Writing, Beamer, Document Formatting.
- NepaTronix 3-Day Drone Training — May 2–4, 2026. Backed by Nepal Police / Drone Association of Nepal / IITM Pravartak. Skills: Drone Technology, UAV Operations, Flight Safety Protocols.
- IEEE SPAx — Engineer Your Profile (May 23, 2026) — Organizing Team Member, IEEE KEC KTM Student Branch. Recognized for outstanding contribution to planning, coordination, and execution. Certificate of Organization received. Skills: Event Management, Leadership, Student Outreach.
- Prompt Engineering Fundamentals — TechAxis (May 8, 2026). 2-hour hands-on workshop. Certificate of Completion. Skills: Prompt Engineering, LLM Interaction, AI Tools.
- IEEE Conference Leadership Workshop 2026 — IEEE Nepal Section. Skills: Event Management, Leadership, Conference Organization.
- IEEE Day 2025 — Organizer (Oct 9, 2025) — IEEE KEC KTM Student Branch. Certificate of Appreciation for dedication and contribution to event planning and execution. Skills: Event Coordination, Leadership.
- IEEEXtreme 19.0 Participant (Team: ShadowXTREME) — Oct 25, 2025. Skills: Competitive Programming, Algorithms, Data Structures, C++.
- Agile Workshop — IEEE KEC KTM Student Branch (Jul 22, 2025). Certificate of Appreciation. Skills: Agile Methodology, Scrum, Project Management.
- PCB Design & Fabrication Workshop — KEC Robotics Club (Jul 24–26, 2025). Three-day hands-on workshop on PCB design principles and fabrication techniques. Certificate of Completion. Skills: PCB Design, Hardware Prototyping, EDA Tools.
- Mentor — Electronics For All Workshop (Jul 27, 2025) — Shree Kuleshwor Secondary School. Mentored school students in foundational electronics. Certificate of Appreciation received.
- How Hackers Bypass Security: A Beginner’s Guide — Offenso Webinar (Jul 2025). Offensive security fundamentals and attack bypass techniques. Certificate of Participation. Skills: Cybersecurity, Offensive Security.
- AWS Cloud Computing Workshop — KEC IT Club (2025). Hands-on with EC2, S3, and cloud deployment fundamentals.
- PreXtreme Competitive Programming Workshop — IEEE Pulchowk (2025).
- Git & GitHub Workshop — KEC IT Club (2024).
- HTML & CSS Workshop — Design, Code & Launch via GitHub Pages (2024) — Microsoft Learn Student Ambassador in association with DPL Teens. Certificate of Completion. Skills: HTML, CSS, GitHub Pages, Web Deployment.

Active member: IEEE KEC KTM Student Branch · Electronics Project Club · KEC Maker’s Space · KEC Music Club

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